dependent only on the prompt neutrons. Such a reactor is said to be *prompt critical*. Nuclear explosives are designed to be prompt critical with a $\Delta k \geq 1$ giving a $t_{per} \leq 10^{-8}$ s, while reactors for power production (thermal as well as fast reactors) should for safety reasons always be operated in the *delayed critical* region. If in a thermal uranium reactor Δk is made $\geq \beta$ ($k_{eff} \geq 1.0065$), t_{per} becomes 13 s for $\theta_{eff} = 0.084$, and (19.34) shows that the neutron amount (and power) doubles in about 10 s. Usually k_{eff} is made smaller than 1.0065 and the power doubling time is correspondingly larger.

Some of the isobar chains from fission contain isotopes with very high n,γ cross sections. One of the most important is $A = 135$ and it's effect on reactor kinetics is used as an example. The genetic relations in the A 135 chain is:

$$\begin{array}{l} \text{fission yield\% } 3.5 \qquad 2.5 \qquad\quad 0.6 \\ \qquad\qquad\quad \downarrow \qquad\quad \downarrow \qquad\quad\ \downarrow \\ {}^{135}\text{Te} \xrightarrow{18\text{s}} {}^{135}\text{I} \xrightarrow{6.61\text{ h}} {}^{135}\text{Xe} \xrightarrow{9.10\text{ h}} {}^{135}\text{Cs} \xrightarrow{2\text{ My}} {}^{135}\text{Ba(stable)} \\ \qquad\qquad\qquad\quad \text{n},\gamma \downarrow\ 2.65\times10^{6}\text{ b} \\ \qquad\qquad\qquad\quad {}^{136}\text{Xe(stable)} \\ \qquad\qquad\qquad\quad \text{n},\gamma \downarrow\ 0.16\text{ b} \end{array} \tag{19.36}$$

During operation at a constant neutron flux the large $\sigma_{n,\gamma}$ continuously converts ^{135}Xe to ^{136}Xe, thereby keeping the amount small and limiting the poisoning effect. However, after a strong reduction in neutron flux ^{135}Xe formation is more rapid by decay of its precursor, ^{135}I, than its destruction, resulting in a transient radioactive equilibrium. As a result the reactivity drops for some hours because of an increasing amount of ^{135}Xe. If the reactor has been shut down for a while, it may be impossible to start it until the amount of ^{135}Xe has been reduced again by decay (*xenon poisoning*). However, if the reactor can be started it burns ^{135}Xe and Δk increases rapidly until a new equilibrium concentration is reached. This is called a *xenon transient* and can temporarily make the reactor difficult to control; if mishandled it may even cause prompt criticality. Once xenon poisoning has become very important after a reactor stop it is practice to delay restart until ^{135}Xe has decreased to a safe level.

19.9. FUEL UTILIZATION

Figure 19.7 shows the consumption of fissile ^{235}U while new fissile ^{239}Pu and ^{241}Pu (as well as some fission products and other actinides) are produced through radiative capture in fertile ^{238}U and ^{240}Pu. The Figure relates to a particular reactor type, and different reactors give somewhat different curves. Figure 19.5 shows the different capture and decay reactions. In §19.2 it is concluded that in a reactor operating at a power of 3 GW_{th} 3.16 kg ^{235}U d^{-1} would be fissioned. Because of the ^{235}U(n,γ)^{236}U reaction, some additional uranium is consumed (19.4). To account for this, we use

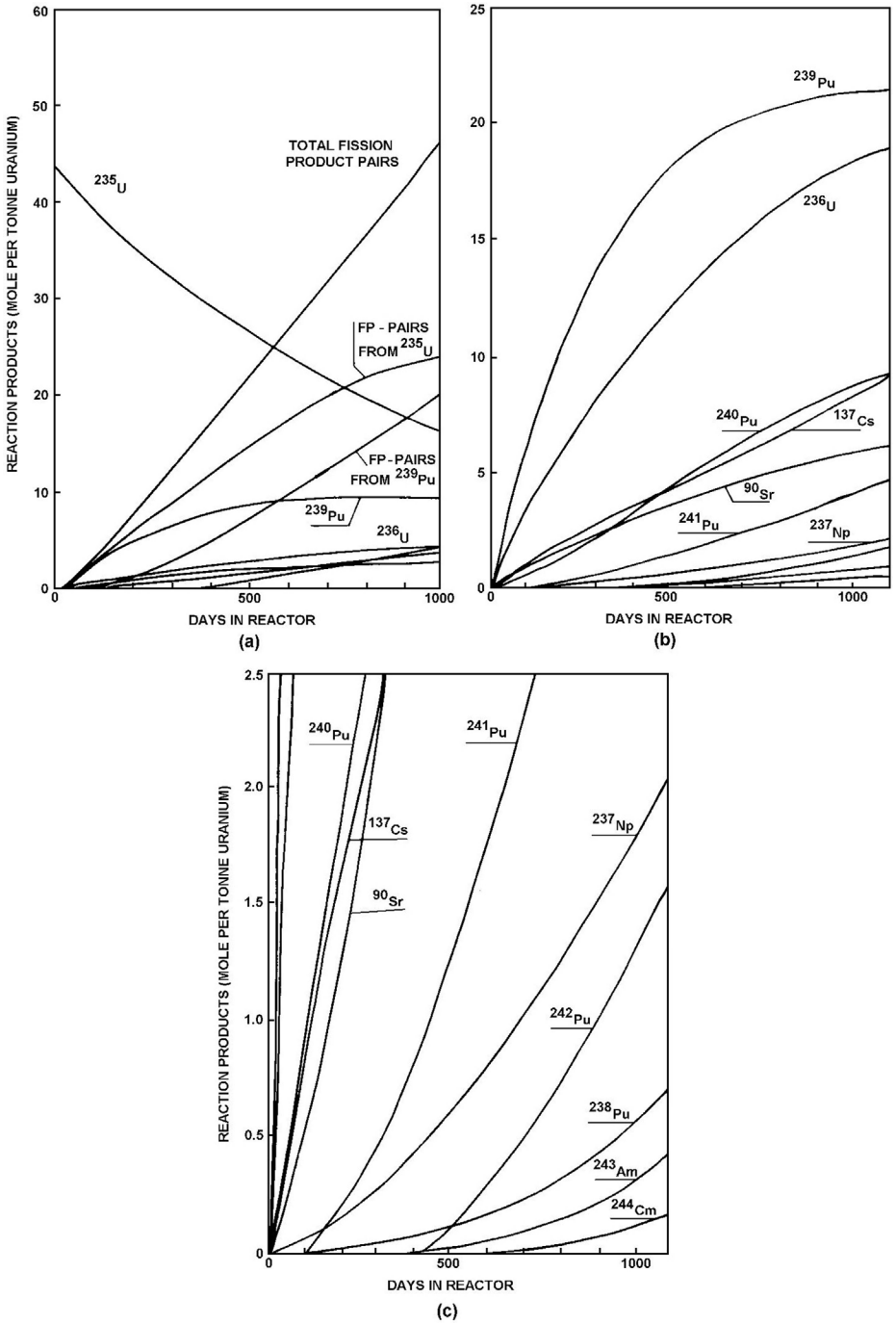


Figure 19.7 Build-up of reaction products in fuel originally enriched in ^{235}U to (a) 1% used in a graphite reactor ($\sim$230 MW_{th}), (b) and (c) 3.3% used in a LWR ($\sim$1000 MW_e).

$(\sigma_f + \sigma_\gamma)/\sigma_f = 1.17$; thus the ^{235}U consumption is $3.16 \times 1.17 = 3.70$ kg d^{-1}. This value holds for a fresh reactor core; however, after a core has operated for some time, fission in ^{239}Pu formed from neutron capture in ^{238}U begins to contribute to the energy production, and, later, fission from ^{241}Pu also contributes. Therefore the ^{235}U consumption rate successively decreases for the same power production in an ageing core. During the whole lifetime of a 2–3% enriched uranium core about 40% of the total energy production comes from fissioning of plutonium isotopes (Fig. 19.8). On the average a 1000 MW_e (33% efficiency) LWR power station consumes daily 2.2 kg ^{235}U and 2.0 kg ^{238}U. About 3.1 kg fission products are produced each day from the fissioning of uranium and plutonium. To calculate these yields, effective cross sections must be used like those given in Figure 19.5. The more ^{239}Pu and ^{241}Pu that is produced for each atom of ^{235}U consumed, the more efficient can be the utilization of fuel. This is expressed by the conversion factor C, which is given by

$$C = \eta - 1 - S \tag{19.37}$$

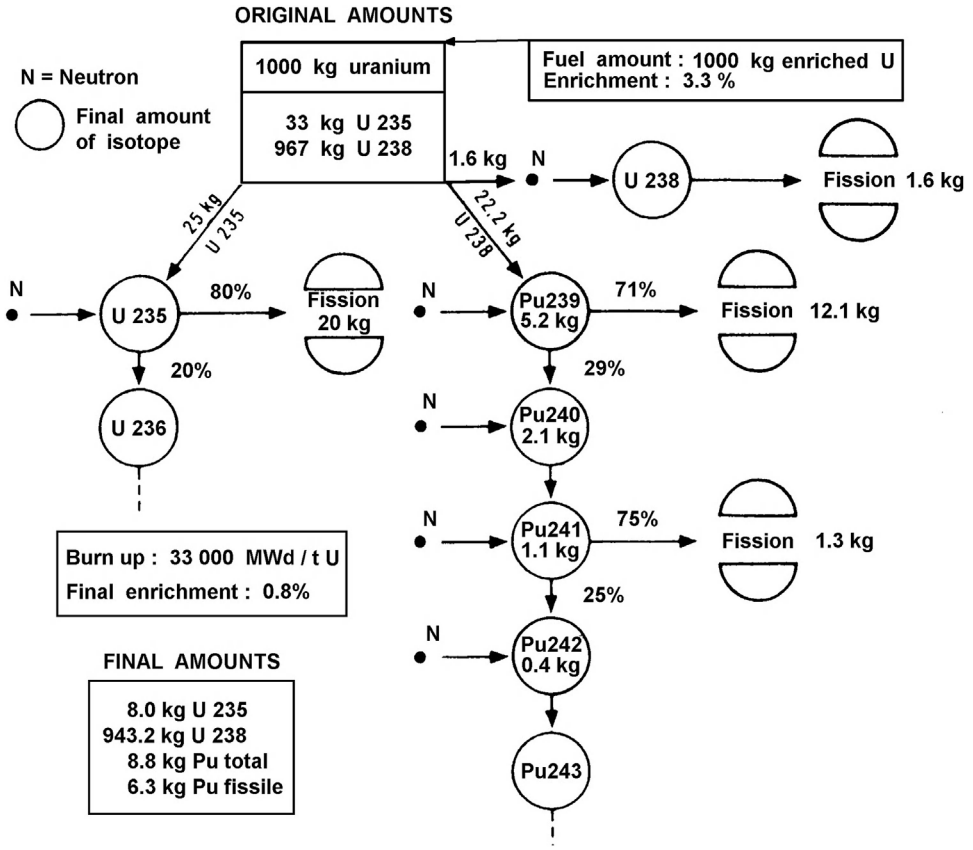


Figure 19.8 Fission product balance in a light water power reactor. *(From F. Hoop.)*

where S is the neutron losses by processes other than fission and capture reactions to produce fissile atoms in the fuel. If we assume $C = 0.8$, which is a valid figure for a heavy water moderated reactor, fission of 100 atoms of ^{235}U leads to the formation of 80 new atoms of ^{239}Pu and ^{241}Pu.

In the second generation one produces C^2 new ^{239}Pu and ^{241}Pu atoms. The fraction of fissile atoms in the fuel which is transformed is $x_i\,(1 + C + C^2 + ...) = x_i/(1 - C)$, where x_i is the original atomic fraction of fissile atoms (0.0072 for natural uranium). The ratio x_i / $(1 - C)$ expresses the maximum utilization of the fuel. If a reactor could be operated so that all the fissionable material, both original and produced in operation, were consumed, it would be possible, assuming $C = 0.8$, to obtain five times greater power production than would be provided by the original concentration of ^{235}U only. Because of the high neutron capture cross section of the fission products, unfortunately this is not possible without reprocessing between cycles. Reactors with values of C close to unity are called *converter reactors* while reactors with C considerably less than 1 are referred to as *burners*. Light water reactors usually have $C \leq 0.7$ and are classified as burners, however, designs with a reduced moderator/fuel ratio acting as converters or thermal breeders have been suggested. Figure 19.9 shows the possible fuel utilization in various reactor types.

The lowering of f (eqn. (19.14)) due to the buildup of fission products and decrease in amount of fissile atoms are the main reasons for fuel replacement. It is obvious that if Σ_a (fuel) is very large, as is the case for highly enriched fuel, higher amounts of fission products can be tolerated; i.e. more energy can be produced from the fuel before f becomes too small.

The fuel utilization is referred to as *burnup*. The burnup may be expressed as the percentage of fuel used before it must be replaced. For example, 1% burnup means that for each ton of fuel 10 kg of the fissile plus fertile atoms have been consumed (in fission *and* capture). However, usually the fuel burnup is given in amount of energy obtained per ton of initially present fuel atoms (in case of mixed U − Pu fuels per ton of initial heavy metal, IHM). The production of 33 000 MWd of thermal energy ($MW_{th}d$) from 1000 kg enriched uranium in a LWR consumes 25 kg ^{235}U and 23.8 kg ^{238}U (Fig. 19.8), or 4.88% of the original uranium amount. Then 1% burnup corresponds to 33 000/4.88 = 6762 $MW_{th}d$/t U. However, if credit is made for the 6.3 kg fissile plutonium formed (which "replaces" the ^{235}U), the total consumption is reduced to 48.8 − 6.3 = 42.5 kg (or 4.25%); a 1% burnup is then equal to 7765 $MW_{th}d$/t U.

Considering the amount of total consumed uranium (∼7 t natural uranium for each ton enriched to 3.3% in ^{235}U), the utilization of natural uranium is much less; if plutonium is recycled, utilization increases to the overall figures given in Figure 19.9.

The maximum conversion factor at the start of a uranium fueled reactor is given by

$$C = \eta\,\varepsilon(1 - p)\,e^{-B^2\tau} + \Sigma_a\,(^{238}\mathrm{U})/\Sigma_a\,(^{235}\mathrm{U}) \tag{19.38}$$

where the two terms give the number of neutrons absorbed by ^{238}U in the resonance absorption region and in the thermal region, respectively, per neutron absorbed in ^{235}U;

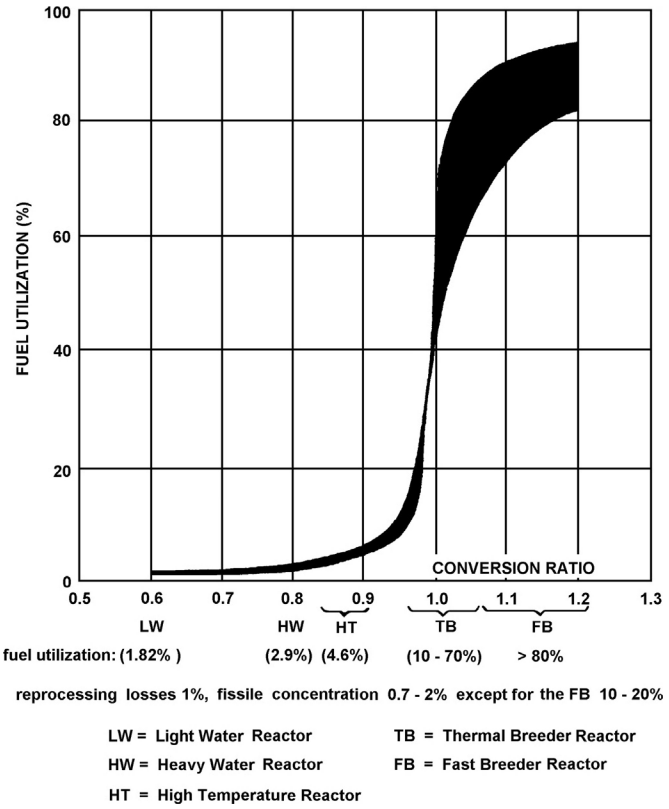


Figure 19.9 Conversion ratios and fuel utilization efficiencies. *(From Thermal Breeder Consultants Group, Salzburg, 1977.)*

the η value refers to ^{235}U. The term $e^{-B^2\tau}$ is the fraction of fast neutrons which does not escape from the reactor (19.16). Using this approximate expression with the reactor example in Table 19.4, we obtain $C = 0.50$. For a reactor core at equilibrium corrections must be introduced for fissions in the plutonium formed; this gives a higher conversion ratio, usually $\sim$0.7 in LWR.

If $C > 1$ the possibility exists of producing more fissile material than is consumed, which is referred to as *breeding*. The breeding ability of a reactor is given by the *breeding gain*, $G = \eta - 2 - S$. Table 19.2 shows that ^{233}U, ^{235}U, ^{239}Pu, and ^{241}Pu all have $\eta > 2$; in practice η is even larger when using effective cross sections (Table 19.2 and Fig. 19.5). In a reactor with these nuclides and fertile material such as ^{232}Th and ^{238}U present it is possible to produce more fissile material than is consumed provided that neutron losses are small; such reactors are called *breeder reactors* or simply *breeders*. While the utilization of the fuel can be increased to perhaps a maximum of 10% with converters, breeding makes it possible to burn $\sim$70% of the fuel material (natural uranium and natural thorium, Fig. 19.9), provided reprocessing and recycling is utilized.

From Table 19.2 it can be seen that the highest η value is exhibited by ^{233}U for thermal neutron energies and by ^{239}Pu for fast neutron energies. This suggests two different types of breeder reactor, the *thermal breeder* based on the reaction $^{232}\text{Th} \rightarrow {}^{233}\text{U}$ and the *fast breeder* based on the reaction $^{238}\text{U} \rightarrow {}^{239}\text{Pu}$. Due to the smaller effective cross sections, these breeders must be charged with a fuel having high concentrations (15−30%) of ^{233}U or Pu. These materials can be produced in conventional non-breeder reactors. Because a single inelastic scattering in ^{235}U may reduce neutron energy considerably, a fast breeder should only contain little of this isotope and hence depleted uranium is preferred over natural uranium as fertile material.

For a large fast breeder reactor like the French Creys-Malville (in operation from 1986 to 1997), the fuel utilization is higher than in the LWRs; a 1% burnup corresponds to about 8500 $\text{MW}_{\text{th}}\text{d t}^{-1}$ IHM. Burnup figures as high as 15% (130 000 $\text{MW}_{\text{th}}\text{d t}^{-1}$ IHM) have been achieved in fast breeders. Breeder reactors are described in Chapter 20.

19.10. THE OKLO PHENOMENON

At the Oklo mine in Gabon uranium has been mined for many years and delivered to the Pierrelatte isotope enrichment plant in France. The ore body, which is estimated to contain 400 000 t of uranium, is quite inhomogeneous, and pockets very rich in uranium are found embedded in sandstone and granite, see Figure 19.10. These pockets are often shaped like a lens, 10 m long and about 1 m in diameter, and contain on the average 10−20% pitchblende, although uranium concentrations up to 85% (pure pitchblende) are found in some spots. In 1972 it was discovered that the isotopic composition of some of the uranium received in France deviated from uranium from other sources. The ^{235}U content was significantly lower than the natural 0.72%. Careful analysis showed that some deliveries contained <0.5% ^{235}U, a serious disadvantage in materials to be used for ^{235}U enrichment. Analysis at Oklo showed samples even lower in ^{235}U content, as well as other elements whose isotopic composition considerably deviated from the natural one. For example, natural neodymium contains 27% ^{142}Nd, while Oklo neodymium contained <2%, see Fig. 19.10 and 19.11. On the other hand, natural neodymium contains 12% ^{143}Nd, while in Oklo, samples containing up to 24% ^{143}Nd were found. Fission product neodymium contains about 29% ^{143}Nd, while ^{142}Nd is not produced by fission. This should be compared with the lack of ^{142}Nd and the excess of ^{143}Nd in the Oklo samples, a condition which was found to be directly related to a high total concentration of uranium but a deficiency of ^{235}U. The conclusion was obvious: in some ancient time the missing ^{235}U had fissioned, producing ^{143}Nd among other fission products. This conclusion was supported by similar investigations on the isotopic composition of other elements produced by fission.

Because ^{235}U has a shorter half-life than ^{238}U, all uranium ores were richer in ^{235}U in the past. From $^{87}\text{Rb}-{}^{87}\text{Sr}$ analysis the age of the Oklo deposit is known to be 1.74×10^9 y; at